

UNIT 3

1) Write short note on Network Address Translation.

Network Address Translation (NAT)

Definition:

Network Address Translation (NAT) is a technique used in computer networking to map private (internal) IP addresses to a public (external) IP address, enabling devices within a private network to communicate with external networks like the Internet.

Purpose:

1. **IP Address Conservation:** NAT allows multiple devices on a private network to share a single public IP address, helping to conserve IPv4 addresses.
2. **Security:** By hiding internal IP addresses, NAT adds a layer of security, as external users cannot directly access internal devices.
3. **Network Flexibility:** Internal networks can use any private IP address scheme without conflicting with public IP addresses.

How it Works:

- A NAT-enabled router modifies the source IP address of outgoing packets from a private IP to its own public IP.
- When a response comes back, NAT translates the destination IP from the public address back to the correct private IP, delivering the packet to the intended internal device.

Types of NAT:

1. **Static NAT:** One-to-one mapping between a private and a public IP address.
2. **Dynamic NAT:** Maps private IPs to a pool of public IPs on a first-come-first-serve basis.
3. **Port Address Translation (PAT) / NAT Overload:** Multiple private IPs share a single public IP using different port numbers.

Example:

- Private IP: 192.168.1.10 → Public IP: 203.0.113.5
- Internal device sends a request to the internet → NAT changes source IP to 203.0.113.5
- Response returns to 203.0.113.5 → NAT forwards it to 192.168.1.10

2) Explain the concept of Class Full and Class less Addressing

Classful Addressing

Saved memory full ⓘ

Definition:

Classful addressing is an **old IP addressing method** in which the **IP address space is divided into fixed classes (A, B, C, D, and E)** based on predefined **network and host portions**.

Key Points:

- IP address = **Network ID + Host ID**
- The division is determined by **fixed boundary bits**.
- No concept of subnet mask was used; the class itself defined how many bits were for the network and how many for hosts.

Class	Starting Bits	Range of First Octet	Default Subnet Mask	No. of Networks	Hosts per Network
A	0	1 – 126	255.0.0.0	128	~ 16 million
B	10	128 – 191	255.255.0.0	16,384	~ 65,000
C	110	192 – 223	255.255.255.0	2 million	254
D	1110	224 – 239	—	Multicasting	—
E	1111	240 – 255	—	Experimental	—

Example:

192.168.1.10 → Class C (default mask: 255.255.255.0)

Limitations:

- Wastage of IP addresses (fixed network size).
- No flexibility in dividing networks.

Classless Addressing (CIDR – Classless Inter-Domain Routing)

Definition:

Classless addressing **removes the fixed class boundaries** and allows the network and host portions to be defined **dynamically** using a **subnet mask** (written as `/n`).

Example:

192.168.1.0/26 → means first **26 bits** are network bits, remaining **6 bits** are for hosts.

Advantages:

- Efficient use of IP addresses (no wastage).
- Allows subnetting and supernetting.
- Improves routing efficiency.

Example:

In classless addressing, you can create subnets like:

192.168.1.0/26 → 64 IPs

192.168.1.64/26 → 64 IPs

instead of being forced to use one large Class C block of 254 IPs.

In Short:

Feature	Classful Addressing	Classless Addressing (CIDR)
Based on	Fixed Classes (A–E)	Variable Length Prefix
Subnet Mask	Default, based on class	User-defined (/n notation)
Flexibility	Rigid	Flexible
IP Utilization	Often wasteful	Efficient
Example	192.168.1.1 (Class C)	192.168.1.1/26

3) Compare Static and Dynamic Routing

1. Static Routing

Definition:

Static routing means the **routes are manually configured** by the network administrator. The router uses **fixed paths** to send data packets.

Features:

- Routes **do not change automatically** even if the network topology changes.
- Suitable for **small networks** with few routers.
- Requires **manual updates** when network changes occur.

Advantages:

- Simple to configure for small networks.
- No extra CPU or bandwidth usage for route calculations.
- More **secure** (no automatic updates from outside sources).

Disadvantages:

- Not scalable for large networks.
- Manual reconfiguration is needed if any link fails.
- Difficult to manage in complex topologies.

2. Dynamic Routing

Definition:

Dynamic routing uses **routing protocols** (like RIP, OSPF, or EIGRP) to **automatically learn and update** routes based on network changes.

Features:

- Routes are **automatically updated** when the network topology changes.
- Suitable for **large and complex networks**.
- Uses algorithms to determine the best path.

Advantages:

- **Automatic adaptation** to network changes or failures.
- **Scalable** and easier to manage in large networks.
- Provides **redundancy** and **fault tolerance**.

Disadvantages:

- Requires more CPU, memory, and bandwidth.
- More complex to configure initially.
- Slightly less secure if not properly managed.

3. Difference Table

Aspect	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Adaptability	Does not adapt to changes	Adapts automatically
Protocols Used	None	RIP, OSPF, EIGRP, BGP, etc.
Overhead	Low	High (due to updates and calculations)
Scalability	Poor	Excellent
Security	More secure	Comparatively less secure
Best Suited For	Small networks	Large networks
Example	Manually adding route using <code>ip route</code> command	Router learns routes using RIP or OSPF

4) Describe in short, the importance and working of ARP Protocol, what is ARP cache.

ARP (Address Resolution Protocol)

1. Importance:

- ARP is used to **find the physical (MAC) address** of a device from its **IP address** within a local network.
 - It enables communication between devices in the same LAN by mapping **IP addresses to MAC addresses**.
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2. Working of ARP:

1. When a device wants to send data, it checks its **ARP cache** to see if it already knows the MAC address of the target IP.
 2. If not found, it **broadcasts an ARP Request** asking, "Who has IP x.x.x.x?"
 3. The device with that IP replies with an **ARP Reply** containing its MAC address.
 4. The sender then stores this IP-MAC mapping in its **ARP cache** for future use.
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3. ARP Cache:

- The **ARP cache** is a **temporary table** stored in a device (like a router or computer) that keeps **recently resolved IP-to-MAC address mappings**.
 - It helps to **speed up communication** by avoiding repeated ARP requests for the same IP address.
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✅ In short:

ARP translates IP addresses to MAC addresses for local communication. The ARP cache stores these mappings temporarily to make the process faster.

5) Differentiate between circuit switching, Packet switching, message Switching.

1. Circuit Switching

Definition:

A **dedicated communication path (circuit)** is established between sender and receiver for the entire duration of the communication.

Key Points:

- The path remains reserved until communication ends.
- Commonly used in **telephone networks**.
- Data is transmitted **continuously** and in order.

Advantages:

- Reliable and ordered delivery.
- Constant transmission rate (no delay once circuit is set).

Disadvantages:

- Wastes bandwidth when the line is idle.
- Setup time is required before transmission.

Example:

Traditional landline phone call.

2. Packet Switching

Definition:

Data is divided into **small packets**, and each packet is sent **independently** through the network using the best available path.

Key Points:

- No dedicated path; packets may take **different routes**.
- Each packet has a **header** with destination information.
- Used in **computer networks and the Internet**.

Advantages:

- Efficient use of bandwidth.
- Fault tolerant — if one path fails, packets can take another route.

Disadvantages:

- Packets may arrive **out of order** or delayed.
- Requires reassembly at the receiver's end.

Example:

Internet data transfer (emails, web browsing, etc.).

3. Message Switching

Definition:

Entire **messages are treated as single units** and stored at intermediate nodes before being forwarded ("store and forward" technique).

Key Points:

- No dedicated path.
- Each message is temporarily stored until the next link is free.
- Requires more storage at intermediate nodes.

Advantages:

- No need for dedicated path.
- Useful for **non-real-time** applications (like telegraphy).

Disadvantages:

- High delay due to store-and-forward process.
- Not suitable for real-time communication.

Example:

Older telegraph and email systems.

4. Difference Table

Feature	Circuit Switching	Packet Switching	Message Switching
Path	Dedicated path	Dynamic paths	No fixed path
Data Unit	Continuous data stream	Packets	Whole message
Resource Usage	Reserved for session	Shared among users	Shared, store-and-forward
Delay	Setup delay only	Transmission delay per packet	High delay (store-and-forward)
Efficiency	Low (idle time wasted)	High	Moderate
Use Case	Telephone network	Internet	Telegraph, email
Order of Delivery	In order	May vary	In order

6) Compare routing protocols RIP, OSPF, BGP

1. RIP (Routing Information Protocol)

Type:

→ Distance Vector Routing Protocol

Key Features:

- Uses **hop count** as the metric (max 15 hops).
- Periodically sends the **entire routing table** to neighbors.
- Simple and suitable for **small networks**.
- **Routing updates** every 30 seconds.

Advantages:

- Easy to configure and manage.
- Low resource usage.

Disadvantages:

- Slow convergence.
- Not scalable for large networks.
- Hop limit (15) restricts network size.

2. OSPF (Open Shortest Path First)

Type:

→ Link State Routing Protocol

Key Features:

- Uses **cost (based on bandwidth)** as the metric.
- Each router builds a **complete map of the network** using LSAs (Link State Advertisements).
- Supports **hierarchical design** using areas.
- **Fast convergence** and supports **VLSM/CIDR**.

Advantages:

- Efficient and scalable.
- Supports large and complex networks.
- Updates are only sent when topology changes (not periodic).

Disadvantages:

- More complex to configure.
- Higher memory and CPU usage.

3. BGP (Border Gateway Protocol)

Type:

→ Path Vector Routing Protocol

Key Features:

- Used for routing between **different autonomous systems (AS)** — i.e., **inter-domain routing**.
- Uses **path attributes** (AS path, policy, etc.) to choose the best route.
- Basis of **Internet routing**.
- Provides **policy-based routing** control.

Advantages:

- Highly scalable (used by ISPs and on the Internet).
- Allows control over routing policies between organizations.
- Very stable once routes are established.

Disadvantages:

- Very complex configuration.
- Slower convergence compared to OSPF.

4. Comparison Table

Feature	RIP	OSPF	BGP
Type	Distance Vector	Link State	Path Vector
Metric Used	Hop Count	Cost (Bandwidth)	Path Attributes
Max Hop Count	15	No limit	No limit
Convergence Speed	Slow	Fast	Moderate to Slow
Scalability	Small networks	Large enterprise networks	Very large (Internet)
Update Type	Periodic	On change	Incremental
Used For	Intra-domain	Intra-domain	Inter-domain
Protocol Number	UDP Port 520	IP Protocol 89	TCP Port 179
Complexity	Simple	Moderate	High
Example Use	Small LAN	Enterprise LAN/WAN	Internet backbone

7) Explain the concept of connection less and connection oriented protocol with example.

1. Connection-Oriented Protocol

Definition:

A **connection-oriented protocol** establishes a **dedicated connection** between sender and receiver before data transmission begins.

It ensures **reliable communication**, where data is delivered **in order** and **without errors**.

Key Features:

- Connection setup, data transfer, and termination phases.
- Acknowledgements are used to confirm receipt.
- Retransmission occurs if packets are lost.
- Provides **error checking**, **flow control**, and **reliability**.

Example:

- **TCP (Transmission Control Protocol)**
 - Used in web browsing (HTTP), email (SMTP), and file transfer (FTP).
 - Example: When you open a website, TCP establishes a connection between your device and the web server using a **three-way handshake**.

Analogy:

Like a **telephone call** — connection is established before conversation.

2. Connectionless Protocol

Definition:

A **connectionless protocol** sends data **without establishing a dedicated connection**.

Each data packet is treated **independently**, and delivery is **not guaranteed**.

Key Features:

- No connection setup or termination.
- Faster but **less reliable**.
- No acknowledgements or retransmissions.
- Suitable for applications where **speed is more important than reliability**.

Example:

- **UDP (User Datagram Protocol)**
 - Used in live streaming, online gaming, and voice/video calls.
 - Example: In a video call, if a few packets are lost, the conversation continues without delay.

Analogy:

Like sending **letters by post** — each is sent individually without confirming delivery.

3. Difference Table

Feature	Connection-Oriented Protocol (TCP)	Connectionless Protocol (UDP)	
Connection Setup	Required (3-way handshake)	Not required	
Reliability	Reliable (acknowledgements, retransmission)	Unreliable (no confirmation)	
Data Delivery Order	In sequence	May arrive out of order	
Speed	Slower	Faster	
Overhead	High	Low	
Example	TCP (HTTP, FTP, SMTP)	UDP (DNS, VoIP, Streaming)	
Analogy	Telephone call	Postal letters	

8) Give the different functions of the Network Layer.

Network Layer (Layer 3 of OSI Model)

Definition:

The **Network Layer** is responsible for **data delivery from source to destination across multiple networks**.

It deals with **logical addressing, routing, and path determination**.

Main Functions of the Network Layer

1. Logical Addressing:

- Assigns **unique IP addresses** to devices.
- Ensures each packet can be identified and delivered to the correct destination network.

2. Routing:

- Determines the **best path** for data to travel from source to destination.
- Uses **routing algorithms and protocols** (e.g., RIP, OSPF, BGP).

3. Packet Forwarding:

- Forwards data packets from one network to another based on routing tables.

4. Fragmentation and Reassembly:

- If a packet is too large for the underlying network, it is **divided (fragmented)**.
- At the destination, fragments are **reassembled** into the original packet.

5. Error Handling and Diagnostics:

- Detects and reports network-level errors (e.g., unreachable host).
- Uses tools like **ICMP (Internet Control Message Protocol)** for reporting.

6. Internetworking:

- Allows communication between **different types of networks** (LANs, WANs) by providing a universal addressing scheme.

7. Congestion Control (optional):

- Helps manage network traffic to **avoid congestion** and packet loss.

Summary Table

Function	Description
Logical Addressing	Assign IP addresses to identify devices
Routing	Selects best path for data
Packet Forwarding	Moves packets toward destination
Fragmentation & Reassembly	Breaks/rejoins packets for transmission
Error Handling	Reports errors via ICMP
Internetworking	Connects different networks
Congestion Control	Manages traffic flow to prevent overload

9) Explain in detail, any two network layer protocols.

1. Internet Protocol (IP)

Definition:

The **Internet Protocol (IP)** is the **primary protocol of the Network Layer** responsible for delivering packets from the source to the destination across interconnected networks.

It provides **logical addressing** and **routing** functions.

Functions of IP:

1. Logical Addressing:

- Every device on a network is assigned a unique **IP address** (e.g., 192.168.1.5).
- This helps identify the **source and destination** of each packet.

2. Packet Routing:

- IP determines the **best path** for data to travel using routing tables.
- Routers use the **destination IP address** to forward packets to the next hop.

3. Fragmentation and Reassembly:

- If a packet is too large for a network, IP divides it into **smaller fragments**.
- The destination host **reassembles** the fragments into the original message.

4. Encapsulation:

- IP encapsulates **Transport Layer segments (TCP/UDP)** into **IP packets** before sending them to the Data Link Layer.

5. Error Handling:

- Works with **ICMP (Internet Control Message Protocol)** to report transmission errors (like unreachable hosts).

Versions of IP:

Version	Full Form	Address Size	Example
IPv4	Internet Protocol version 4	32-bit	192.168.1.10
IPv6	Internet Protocol version 6	128-bit	2001:0db8::1

Example:

When a user visits a website:

Your computer's IP (source) → sends a request packet to the web server's IP (destination) using routers that forward the packet along the best path.

2. Internet Control Message Protocol (ICMP)

Definition:

The **Internet Control Message Protocol (ICMP)** is used by network devices (like routers) to **send error messages and operational information** about network communication.

Functions of ICMP:

1. Error Reporting:

- Reports problems like:
 - Destination unreachable
 - Time exceeded (packet TTL expired)
 - Source quench (network congestion)

2. Network Diagnostics:

- Used by tools like **ping** and **traceroute** to check connectivity and route paths.
- Example:
 - `ping` → Tests if a host is reachable.
 - `traceroute` → Shows the path packets take across routers.

3. Flow Control Assistance:

- Helps routers and hosts manage network congestion by notifying when data cannot be delivered.

ICMP Message Types:

Type	Message Name	Purpose
0	Echo Reply	Response to ping
3	Destination Unreachable	Host/network not reachable
8	Echo Request	Sent by ping to check connectivity
11	Time Exceeded	TTL expired before reaching destination

Example:

If you try to access a server that's offline, your router sends back an **ICMP "Destination Unreachable"** message to notify you that the server cannot be reached.

10) What is Routing? Explain the different routing protocols

(REFER QUESTION NO 6 FOR ROUTING PROTOCOLS)

Definition:

Routing is the process of **selecting the best path** for data to travel from the **source** to the **destination** across interconnected networks.

In simple terms — it's how routers decide **where to send packets next** so they reach their final destination.

Functions of Routing:

1. Path Determination:

Chooses the most efficient route based on metrics like hop count, bandwidth, delay, or cost.

2. Packet Forwarding:

Forwards data packets from one router to the next toward the destination.

3. Network Discovery:

Routers learn about other networks and build routing tables using protocols.



Types of Routing

Type	Description	Example
1. Static Routing	Routes are manually set by the administrator.	Small networks
2. Dynamic Routing	Routers automatically learn and update routes using routing protocols.	Large/complex networks
3. Default Routing	Used when no specific route is known; all unknown packets are sent to a default router.	Gateway to Internet

11) Write a short note on: Mobile IP.

Mobile IP

Definition:

Mobile IP is a **network layer protocol** that allows a device (called a *mobile node*) to move from one network to another **without changing its IP address**.

It enables **seamless Internet connectivity** even when the user moves across different networks.

Need for Mobile IP:

In traditional IP, when a device changes its network, its **IP address also changes**, breaking ongoing connections.

Mobile IP solves this problem by keeping the same home IP address regardless of location.

Key Components:

1. Mobile Node (MN):

The device that moves from one network to another (e.g., smartphone, laptop).

2. Home Agent (HA):

A router on the **home network** that maintains the mobile node's permanent IP address and forwards packets to it.

3. Foreign Agent (FA):

A router on the **visited (foreign) network** that helps the mobile node connect and communicate.

4. Care-of Address (CoA):

A temporary IP address assigned to the mobile node while it's on the foreign network.

Working of Mobile IP:

1. When the mobile node moves to a new network, it registers its **Care-of Address** with its **Home Agent**.
 2. The **Home Agent** intercepts packets sent to the mobile node's **home address** and **tunnels** them to the **Care-of Address**.
 3. The **Foreign Agent** receives the tunneled packets and delivers them to the mobile node.
 4. Replies from the mobile node are sent directly to the destination host.
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Advantages:

- Maintains ongoing connections while moving across networks.
- Provides seamless Internet access for mobile devices.
- Supports mobility across different networks and subnets.

Applications:

- Mobile phones and laptops with wireless Internet.
- IoT and vehicular networks.
- Remote work and mobile computing.

12) Explain Link State Routing Algorithm with example

Link State Routing Algorithm

Definition:

Link State Routing (LSR) is a routing algorithm in which **each router has complete knowledge of the network topology** and independently calculates the **shortest path** to every other router using this information.

It is a **Link State Protocol**, used in protocols like **OSPF** and **IS-IS**.

Key Features:

1. Each router **discovers its neighbors** and their link costs.
2. Routers create **Link State Packets (LSPs)** containing information about neighbors and link costs.
3. LSPs are **flooded** to all routers in the network.
4. Each router builds a **complete network map** (topology database).
5. Routers use **Dijkstra's Shortest Path First (SPF) algorithm** to compute the shortest path to each destination.

Steps of the Algorithm:

1. **Discover Neighbors:** Each router identifies directly connected routers and link costs.
2. **Create Link State Packet (LSP):** Contains router ID, neighbors, and cost of each link.
3. **Flood LSPs:** LSPs are sent to all routers in the network so each router has the same information.
4. **Build Topology Table:** Each router constructs a network graph using all received LSPs.
5. **Compute Shortest Path:** Apply **Dijkstra's algorithm** to calculate the shortest path to all destinations.
6. **Update Routing Table:** Based on the shortest paths, routing table entries are created for packet forwarding.

Example:

Consider a network with 4 routers (A, B, C, D) and the following link costs:

Link	Cost
A – B	1
A – C	4
B – C	2
B – D	6
C – D	3

Step 1: Discover Neighbors

- Router A knows B (1) and C (4)
- Router B knows A (1), C (2), D (6)
- Router C knows A (4), B (2), D (3)
- Router D knows B (6), C (3)

Step 2: Create LSPs and Flood

- Each router sends its LSP to all other routers.
- After flooding, all routers have the same network topology.

Step 3: Compute Shortest Path (Dijkstra)

- From **Router A**:
 - Path to B = $A \rightarrow B = 1$
 - Path to C = $A \rightarrow B \rightarrow C = 1 + 2 = 3$ (shorter than direct $A \rightarrow C = 4$)
 - Path to D = $A \rightarrow B \rightarrow C \rightarrow D = 1 + 2 + 3 = 6$

Step 4: Routing Table for A

Destination	Next Hop	Cost
B	B	1
C	B	3
D	B	6

Advantages of Link State Routing:

- Fast convergence compared to Distance Vector Routing.
- Each router has complete knowledge of the network.
- Supports hierarchical networks (areas) and scalable designs.
- Prevents routing loops.

Disadvantages:

- Requires more memory to store network topology.
- Requires more CPU for Dijkstra calculations.
- High overhead for flooding LSPs in large networks.

✅ In short:

Link State Routing allows each router to have a **full map of the network**, computes shortest paths using **Dijkstra's algorithm**, and builds a **routing table** for efficient and loop-free packet delivery.

13) Explain Distance Vector Routing Algorithm with example.

Distance Vector Routing Algorithm

Definition:

Distance Vector Routing (DVR) is a routing algorithm in which each router maintains a **routing table** containing the **distance (cost) to reach each destination** and the **next hop (vector)**.

Routers **periodically share their routing tables with immediate neighbors** to update their knowledge of the network.

Key Protocols: RIP (Routing Information Protocol) uses DVR.

Key Features:

1. Each router knows only about its **directly connected neighbors** initially.
2. Routers exchange **distance vectors** (routing tables) with neighbors periodically.
3. Each router updates its routing table using the **Bellman-Ford equation**:

$$D_x(y) = \min\{c(x, v) + D_v(y)\}$$

Where:

- $D_x(y)$ = Distance from router X to destination Y
 - $c(x, v)$ = Cost to reach neighbor V
 - $D_v(y)$ = Distance from neighbor V to destination Y
4. Routers gradually learn the **shortest path** to all destinations.

Steps of the Algorithm:

1. Initialization:

- Each router sets the distance to itself = 0, and to all other routers = ∞ .

2. Exchange Information:

- Routers send their distance vector to all neighbors periodically.

3. Update Routing Table:

- Each router updates its table based on neighbors' vectors using the formula above.

4. Convergence:

- The process repeats until all routers have consistent shortest path distances.

Example:

Consider a network with 3 routers: A, B, C

Link	Cost
A – B	2
A – C	5
B – C	1

Step 1: Initialization

- Router A: Distance to A=0, B=2, C=5
- Router B: Distance to B=0, A=2, C=1
- Router C: Distance to C=0, A=5, B=1

Step 2: Exchange distance vectors

- A receives B's vector: B→C=1
- Update A→C via B: A→B→C = 2+1=3 (better than 5)
- Update A's routing table: A→C = 3 via B

Step 3: Updated Routing Tables

Router A:

Destination	Next Hop	Cost	
B	B	2	
C	B	3	

Router B:

Destination	Next Hop	Cost
A	A	2
C	C	1

Router C:

Destination	Next Hop	Cost
A	B	3
B	B	1

Step 4: Convergence

- After several updates, all routers have **shortest paths to every destination**.

Advantages of Distance Vector Routing:

- Simple and easy to implement.
- Requires less memory than Link State Routing.
- Suitable for small networks.

Disadvantages:

- Slow convergence, prone to **routing loops**.
- Uses **periodic updates** even when no changes occur.
- Less efficient for large networks.

✅ In short:

Distance Vector Routing relies on routers **sharing their distance tables with neighbors**.

Each router gradually computes the **shortest path to all destinations** using the **Bellman-Ford algorithm**.